

1234 M000 · Variables — English Worksheet (Beginner)

Topic: Variables · **Course:** Builder Basics · **Level:** Beginner · **Time:** about 30 minutes

1 · Predict 🧠

A **variable** is a name that holds a number. `length = 5` means the name `length` now holds 5.

Read the steps. Circle your answer.

```
length = 5
repeat length times:
  place block down
  move forward
```

How many blocks does the agent place? Circle one: 5 · 1 · 10

What does `length` hold? Circle one: a number · a block · a turn

```
length = 3
repeat length times:
  place block down
  move forward
```

How long is the path? Circle one: 3 blocks · 5 blocks · 8 blocks

2 · Find the Difference 🐛

Clean steps first, then a broken version. Circle the bug.

Pair A — The path should be **6 blocks long**.

```
# clean
length = 6
repeat length times:
  place block down
  move forward
```

```
# buggy
repeat length times:
  place block down
  move forward
length = 6
```

What is wrong? Circle one: the number is set too late · the loop is missing · a block is missing

Pair B — The loop should use the name `length`.

```
# clean
length = 7
repeat length times:
  place block down
  move forward
```

```
# buggy
length = 7
repeat size times:
  place block down
  move forward
```

What is wrong? Circle one: the name does not match · the number is wrong · the turn is missing

3 · Fill the Gap

The agent should build a path 4 blocks long. One line is missing.

```
_____  
repeat length times:  
    place block down  
    move forward
```

Which line is missing? Circle one: $\text{length} = 4 \cdot \text{turn right} \cdot \text{move up by } 1$

4 · Show Your Work 📷 🎥

Switch to your homework world. Build a **path** using a variable called `length`. Run it. Then change the number in `length` and run it again.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does. Point at the line that sets your variable.

2 · Your build. Point the camera. Show both sizes you built.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.